MANUEL CORREA GIRALDO

Industrial Organization | Energy Economics | Applied Microeconomics | Electricity Markets
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RESEARCH INTERESTS

Industrial Organization; Energy Economics; Electricity Markets; Price Formation; Renewable Energy Integration; Regulation and Market Design; Long-Term Contracting and Risk in Energy Markets; Applied Microeconomics.

EDUCATION

Ph.D. in Economics 2018 – 2023 (degree conferred 2024)
Universidad EAFIT | Medellín, Colombia
Dissertation: *"Strategic behaviors, pass-through, and the impact of new renewable energy on wholesale cost and retail price in electricity markets: A case study for Colombia"*
Advisor: John J. García-Rendón

M.Sc. in Economics 2015 – 2016
Universidad EAFIT | Medellín, Colombia
Thesis: *"Fiscal Policy and Inequality in a CGE Model for Colombia"* (Honorable Mention)
Advisor: Jesús A. Botero-García

B.Sc. in Mathematics 2005 – 2012
Universidad de Antioquia | Medellín, Colombia
Thesis: "*Radio etiquetado de grafos" [In Spanish. English title: Radio Labeling of Graphs]*.
Advisor: Roberto Cruz Rodes

PUBLICATIONS

Peer-reviewed journal articles

- García-Rendón, J. J., **Correa-Giraldo, M.**, & Gutiérrez-Gómez, A. (2024). "Efecto de la entrada en operación de la central hidroeléctrica más grande y de las energías renovables no convencionales en Colombia sobre el precio de bolsa." *Cuadernos de Economía, 43*(91), 1–24. [In Spanish. English title: *Effect of the start of operation of Colombia's largest hydroelectric plant and of non-conventional renewable energies on the spot price.*] https://doi.org/10.15446/cuad.econ.v43n91.103081
- **Correa-Giraldo, M.**, García-Rendón, J. J., & Pérez, A. (2021). "Strategic behaviors and transfer of wholesale costs to retail prices in the electricity market: Evidence from Colombia." *Energy Economics, 99*, 105276. https://doi.org/10.1016/j.eneco.2021.105276
- Ramírez-Hassan, A., & **Correa-Giraldo, M.** (2019). "Focused estimation for noisy and small data sets: a Bayesian minimum expected loss estimator approach." *Australian & New Zealand Journal of Statistics, 61*(3), 360–379. https://doi.org/10.1111/anzs.12271

Books and book chapters

- García-Rendón, J. J., **Correa-Giraldo, M.**, & Pérez, A. (2023). "Energías renovables no convencionales y efecto orden de mérito en el mercado eléctrico colombiano." In D. Patiño Echeverri & J. García Rendón (Eds.), *Mecanismos de mercado para una transición energética resiliente en Colombia* (pp. 41–66). Bogotá: Editorial Universidad Nacional de Colombia. ISBN 978-958-505-292-5. [In Spanish. English title: *Non-conventional renewable energies and the merit-order effect in the Colombian electricity market.*]
- Perfetti, J. J., Botero, J., Oviedo, S., Forero, D., Higuera, S., **Correa, M.**, & García, J. (2017). *Política comercial agrícola: nivel, costos y efectos de la protección en Colombia*. Bogotá: Fedesarrollo, 207 pp. [In Spanish. English title: *Agricultural trade policy: level, costs and effects of protection in Colombia.*]

Peer-reviewed conference proceedings

- García-Rendón, J. J., **Correa-Giraldo, M.**, & Velásquez-Ceballos, H. (2019). "Retail Electricity Market Price in Colombia." *2019 FISE-IEEE/CIGRE Conference – Living the Energy Transition*. IEEE. https://doi.org/10.1109/FISECIGRE48012.2019.8984955

Conference papers

- Botero García, J. A., **Correa-Giraldo, M**., & García, J. (2018). "The cost of tariff and non-tariff protection in Colombia: A computable general equilibrium exercise." Paper presented at the 21st Annual Conference on Global Economic Analysis, Cartagena, Colombia. GTAP Resource 5580. https://www.gtap.agecon.purdue.edu/resources/res_display.asp?RecordID=5580

WORKING PAPERS

- Patiño Echeverri, D., **Correa Giraldo, M.**, & García-Rendón, J. J. (2019). "Integración de fuentes de energías renovables no convencionales y redes inteligentes en Estados Unidos: evidencia para PJM." Working paper, Repositorio Institucional Universidad EAFIT. [In Spanish. English title: *Integration of non-conventional renewable energy sources and smart grids in the United States: evidence for PJM.*]

CONFERENCE & SEMINAR PRESENTATIONS

2023 - VI Coloquio Nacional de Estudiantes Doctorales en Economía, Universidad Nacional de Colombia, Medellín, Colombia.
"Unveiling the Gap: Assessing the Discrepancy between Wholesale Electricity Price Reductions and Consumer Benefits due to Renewable Energy Integration."

2022 - 24th Applied Economics Meeting, Asociación Libre de Economía (ALdE), Palma de Mallorca, Spain.
"Energías renovables no convencionales y efecto orden de mérito en el mercado eléctrico mayorista en Colombia." [Non-conventional renewable energies and the merit-order effect in the Colombian wholesale electricity market.]

2021 - Seminario Permanente de Economía, Universidad de Antioquia, Medellín, Colombia.
"Strategic Behaviors and Transfer of Wholesale Costs to Retail Prices in the Electricity Market: Evidence from Colombia."

2019 - FISE-IEEE/CIGRE Conference – Living the Energy Transition, Medellín, Colombia.
"Retail Electricity Market Price in Colombia."

2018 - 21st Annual Conference on Global Economic Analysis, Global Trade Analysis Project (GTAP), Cartagena, Colombia.
"Fiscal Policy and Inequality in a CGE Model for Colombia."

2016 - Caribbean Economic Association, Medellín, Colombia.
"Focused Econometric Estimation: The interplay between the Bayesian and frequentist approaches."

TEACHING EXPERIENCE

Adjunct Lecturer | Undergraduate & Graduate Programs 2017; 2019 – present
Universidad EAFIT | Medellín, Colombia

- Taught Data Analysis, Advanced Econometrics, Time Series, Advanced Mathematics (graduate course), and Mathematics at undergraduate and graduate levels; instructor for the Mathematics Bootcamp for Master's and Ph.D. students in Economics.

Adjunct Lecturer 2012 – 2015
Universidad de Antioquia | Medellín, Colombia

- Taught Calculus, Biostatistics, Design of Experiments, and Mathematics.

ACADEMIC & RESEARCH APPOINTMENTS

Visiting Research Fellow Aug. 2022 – Jan. 2023
Duke University – Nicholas School of the Environment | Durham, NC, United States

- Advanced the development and completion of doctoral dissertation research on electricity markets, renewable energy integration, and wholesale-to-retail price transmission.
- Participated in academic seminars and attended an in-person course, broadening my exposure to methodological approaches in energy-related fields.

Research Assistant | Project 6: Policy, Regulation & Markets Jan. 2018 – Dec. 2022
Energética 2030 | Universidad EAFIT / Universidad Nacional de Colombia | Medellín, Colombia

- Developed economic models and electricity-market simulations to evaluate the effects of renewable energy integration on wholesale electricity prices and market outcomes.
- Investigated strategic behavior among electricity-market participants, wholesale-to-retail cost pass-through, and the implications of market structure for electricity price formation.
- Conducted comparative analyses of electricity-market regulation and design, including PJM and CAISO, with emphasis on renewable energy integration and emerging technologies.

PROFESSIONAL & CONSULTING EXPERIENCE

Wholesale Energy Market Professional Jun. 2023 – Aug. 2026
Empresas Públicas de Medellín – EPM | Medellín, Colombia

- Developed quantitative models and scenario analyses to support electricity pricing, commercial-risk assessment, long-term contracting, and the evaluation of investment opportunities.
- Designed pricing methodologies and analytical frameworks for long-term electricity contracting and auctions, including bid-price analysis supporting the award of approximately 700 GWh annually in a Ministry of Mines and Energy auction.

- Led the commercial analysis and structuring of more than ten long-term power purchase agreements (PPAs), associated with approximately 380 MW of generation capacity and 1,000 GWh of annual energy.
- Designed a quantitative model to assess market risk in long-term electricity contracting, integrating generation and sales profiles, energy procurement costs, and alternative contracting levels and pricing methodologies to estimate minimum selling prices.

Consultant | Economic Analysis & Public Policy Oct. 2017 – Feb. 2018
Universidad EAFIT | Medellín, Colombia

- Analyzed census and socioeconomic data to assess the productive structure and economic profile of the Municipality of Envigado, translating quantitative evidence into public-policy recommendations.

Consultant | Agricultural Trade Policy Nov. 2016 – Mar. 2017
Fedesarrollo / Universidad EAFIT | Medellín, Colombia

- Evaluated agricultural trade-policy scenarios using a Computable General Equilibrium (CGE) model, quantifying effects on output, employment, trade, and welfare across income groups.

RESEARCH METHODS & SOFTWARE

Research methods: Econometric analysis; time-series analysis and forecasting; quantitative economic modeling; electricity-market simulation; market design and competition analysis; computable general equilibrium (CGE) model applications; scenario and sensitivity analysis.
Software & programming: Python, R, Stata, MATLAB, GAMS.

AWARDS, SCHOLARSHIPS & RESEARCH FUNDING

- Doctoral Scholarship – Colombia Científica / Energética 2030 2018 – 2022
- Research Scholarship – M.Sc. in Economics, Universidad EAFIT 2016
- Honorable Mention – Master's Thesis in Economics, Universidad EAFIT2016
- Academic Distinction (Highest GPA) – Institute of Mathematics, Universidad de Antioquia 2010

LANGUAGES

- **Spanish:** Native.
- **English:** Advanced (C1), Aptis certificate (2023).

REFERENCES

- **Dalia Patiño-Echeverri** - Gendell Family Associate Professor of Energy Systems and Public Policy, Nicholas School of the Environment, Duke University, Durham, NC, United States. dalia.patino@duke.edu
- **John J. García-Rendón** - Research Professor, Universidad EAFIT, Medellín, Colombia. jgarcia@eafit.edu.co
- **Andrés Ramírez-Hassan** - Distinguished Professor, School of Finance, Economics and Government, Universidad EAFIT, Medellín, Colombia. aramir21@eafit.edu.co